Robotics Assignment: Formulation of DH Transformation from DH Table

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Question 1: Formulate the steps for DH transformation from a DH table.

Answer: To standardize coordinate frames for spatial linkages, the Denavit-Hartenberg (DH) convention uses four kinematic parameters to map the frames:

- **Joint Angle (θ_i):** Rotation about the previous Z_{i-1} axis.
- **Joint Offset (d_i):** Translation along the previous Z_{i-1} axis.
- **Link Length (a_i or r_i):** Translation along the X_i axis.
- **Link Twist (α_i):** Rotation about the X_i axis.

The standard homogeneous transformation matrix from frame $\{i-1\}$ to frame $\{i\}$ is formulated as:

$$T_{i-1}^{i} = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i)\cos(\alpha_i) & \sin(\theta_i)\sin(\alpha_i) & a_i\cos(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) & -\cos(\theta_i)\sin(\alpha_i) & a_i\sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & a_i\sin(\theta_i) \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

To determine the total transformation from the base frame to the end-effector (frame H), multiply the individual link matrices sequentially: ${}^{0}T_H = A_1A_2A_3\ldots$.

Question 2: Assign the necessary frames to the 2-link robot and derive the DH table.

Answer:

Link (#)	θ	d	a	α
0-1	θ_1	0	a_1	0
1-H	θ_2	0	a_2	0

Question 3: Assign the necessary frames to the 3-link robot and derive the forward kinematic equation.

Answer:

Link (#)	θ	d	a	α
0-1	θ_1	0	a_1	0
1-2	$90+\theta_2$	0	0	90
2-H	θ_3	d_3	0	0

The forward kinematics equation for this robot is formulated as ${}^{0}T_H = A_1A_2A_3$.

Question 4: Assign the necessary frames to the 6-DOF robot (all revolute) and derive the DH table.

Answer:

Link (#)	θ	d	a	α
0-1	θ_1	0	0	90
1-2	θ_2	0	a_2	0
2-3	θ_3	0	a_3	0
3-4	θ_4	0	a_4	-90
4-5	θ_5	0	0	90
5-6	θ_6	0	0	0

Question 5: Assign the necessary frames to the 6-DOF robot (with prismatic joint) and derive the DH table.

Answer:

Link (#)	θ	d	a	α
0-1	θ_1	0	0	-90
1-2	θ_2	d_2	0	90
2-3	0	d_3	0	0
3-4	θ_4	0	0	-90
4-5	θ_5	0	0	90
5-6	θ_6	0	0	0

Question 6: Assign the necessary frames to the 4-DOF robot and derive the DH table.

Answer:

Link (#)	θ	d	a	α
0-1	θ_1	0	0	90
1-2	0	l_5	0	90
2-3	0	l_7	0	90
3-H	θ_4	0	0	0